RL pre-lecture hw5

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1 (1)

(a) Neural network dynamic model: Parametric. The model's capacity is completely defined by a fixed number of network weights and biases(θ) predetermined by the architecture. It doesn't store historical data. Instead, it compresses environment transition dynamics into these fixed parameters to approximate $s_{t+1} = f_\theta(s_t, a_t)$.
(b) A replay buffer used as an empirical transition model: non-parametric. The model is represented directly by the collection of transition samples (s, a, r, s') stored in the memory. The model size grow linearly with the amount of experienced data rather than bounded by a fixed set of parameters.
(c) A symbolic PPDL-style planning model: non-parametric. Use symbol to describe the task instead of using sampling data to learn the parameters.
(d) A Gaussian process dynamics model: non-parametric. As a Bayesian non-parametric approach, a Gaussian Process doesn't assume a fixed functional form the dynamics. Instead, the prediction at a new state-action pair depends on its kernel-defined correlation with all accumulated training data points, meaning the underlying covariance matrix grows with the sample size.
(e) A table of all observed transitions in a small grid-world: non-parametric. Tabular representation don't impose any parametric assumptions (such as linearity or neural configurations) on the environment distribution. The lookup table simply aggregates and computes transition frequencies $\frac{N(s,a,s')}{N(s,a)}$ directly from data, making it a classic non-parametric maximum likelihood estimator.

2 (2)

(a) At each time step $t = 0, 1, 2, \ldots, T$
Step 1: Plan the current state s_t for a future horizon $H < T$ (e.g., by random shooting or CEM).
Step 2: Execute the first planned action, observe the next state s_{t+1} and repeat.
(b) The reason why MPC usually execute only the first planned action is 1. The learned model can't be 100 percent accurate. If we execute next H steps at the same time, the tiny predicting error will accumulate by the time. (c) Self-driving car is a real-world sample of using MPC. The car can use model to predict like future 5 seconds driving path, but in every 0.1 second, based on the sensor's information like lidar, it will recalculate the turning of the steering wheel in case of the immediately accident.
(d)

- Plans entirely in a decoder-free latent space, optimizing only task-relevant features.
- Appends a learned Q-function at the end of the short horizon(z_H) to capture long-term expected returns.
- Uses a learned actor π_θ to seed the initial means of the MPPI planner, heavily accelerating continuous action optimization.
- Employs a regularization loss to match predicted latent states with target representations, mitigating compounding drift.
- Jointly trains embedding, dynamics, reward, and value networks end-to-end use TD loss for maximal sample efficiency.